

SUJITH MAKAM

Software Engineer | FullStack AI Developer

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 [Leet Code](#)

Education

Indian Institute of Information Technology (IIIT), Dharwad

DEC 2021 - JULY 2025

B.Tech in Computer Science and Engineering

CGPA: 8.41

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript

Cloud & DevOps: AWS, CDK, AZURE, Docker

Web: ReactJs, NodeJs, ExpressJs

AI/ML: Generative AI, Agentic AI, LLMs

Databases: MongoDB, MySQL

Operating Systems: Ubuntu, Windows.

Experience

CGI

Jul. 2025 - Present

Software Engineer

- Designed and deployed an **Agentic AI-driven orchestration framework** integrating Azure AI, Cognigy, Genesys, and AWS services to automate enterprise workflows, improving decision accuracy and process reliability across customer interaction systems.
- Implemented **end-to-end Infrastructure-as-Code (IaC)** using **AWS CDK (TypeScript)** to provision and deploy complete **Amazon Connect** environments, enabling scalable, automated cloud contact-center deployments.
- Engineered a **production-grade Amazon Connect DR (Disaster Recovery) synchronization solution** using **boto3, AWS SDK, Lambda, EventBridge, and SSM**, delivering automated cross-region configuration sync and ensuring high availability of contact-center operations.
- Architected and implemented a **Generative AI-powered call summarization solution**, reducing post-call handling time and operational costs by streamlining documentation and improving customer service efficiency.
- Developed a **proof-of-concept agentic AI solution** that autonomously coordinates cross-departmental workflows, significantly reducing manual intervention and minimizing SLA breaches.

Indian Institute of Science Bangalore

May. 2024 - July. 2024

Research Intern

- Investigated LLM Pruning techniques for improving model efficiency, optimizing the attention and feed-forward layers of GPT-2. Achieved a 5.2% reduction in instruction cache misses through tiled matrix multiplication implementation.
- Conducted detailed profiling of GPT-2 using Intel VTune, identifying bottlenecks and enhancing cache efficiency, which resulted in a 4x speedup in attention layer execution.
- Optimized identified bottlenecks in the attention and feed-forward layers of GPT-2 and Reduced memory bandwidth usage by 34.8 %, decreasing memory-bound issues and ensuring more efficient execution
- Implemented custom classes (e.g., tiled matrix multiplication) for performance improvement in the attention layer.

Projects

Cloud Notebook



- The Cloud Notebook project utilizes React and Express to deliver a dynamic, user-friendly digital notebook. React ensures an intuitive frontend, Node.js and Express ensure a secure backend.
- Implemented features for real-time note-taking and organization, enhancing productivity and usability.

Lightweight IoT Intrusion Detection System (Edge-AI)



- Developed a scalable Curriculum Learning framework with XAI (LIME) validation, utilizing optimized GRU/LSTM attention layers to progressively detect complex cyber-attacks.
- Attained 98 % accuracy on the CIC-IoV-2024 dataset while reducing model size to 367KB via quantization and pruning, enabling high-efficiency deployment on resource-constrained edge devices.

FAPL-DM-BC: Blockchain-Federated Security Framework



- Designed a privacy-preserving IoV framework merging Federated Learning with Blockchain smart contracts for decentralized model validation and tamper-proof provenance.
- Achieved 93.1% accuracy with 42ms latency by engineering a Dockerized microservices architecture (using SMPC) and integrating a dual-model XAI feedback loop (SHAP/LIME) for real-time threat detection.

Crop Recommendation System



- Developed a crop recommendation system that identifies the optimal crop for the next season based on the environmental factors, enhancing farming decisions.
- Analyzed soil nutrient levels and environmental conditions, leveraging machine learning for precise predictions with a user-friendly input and visualization interface.
- Attained 99% accuracy using the Random Forest algorithm, ensuring highly reliable crop recommendations.

Certifications

- Full Stack Web Development from Udemy.
- Machine Learning A-Z from Udemy.



Research Publications

Research Paper Publication and Presentation: COMSNETS 2026

Published and presented a paper titled "FAPL-DM-BC: Adaptive Privacy-Aware, Blockchain-Backed FL Framework with Dual-Model XAI for the IoVs" at the COMputer COMMunication Systems and NETworkS Forum 2026

Research Paper Publication and Presentation: IEEE FNWF 2025

Published and presented a paper titled "Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI" at the IEEE Future Networks World Forum 2025

Research Paper Publication and Presentation: IEEE FNWF 2025

Published and presented a paper titled FedGuard-IIoT: A Scalable Federated Learning Architecture for Network-IDS in the Industrial IoT" at the IEEE Future Networks World Forum 2025

Research Paper Publication and Presentation: IEEE SPACE 2025

Published and presented a paper titled "Predicting Stellar Metallicity: A Comparative Analysis of Regression Models for Solar Twin Stars" at the IEEE International Conference on Signal Processing, Computing and Control (SPACE) 2025

Achievements

TCS TechBytes: Dharwad Regional Finals Runner Up

A Prestigious Tech Quiz conducted by TCS and BITES in Katakana

State Rank Holder: Andhra Pradesh State Board Topper

Scored 990/1000 in my state board examinations